Building a Production-Quality, Scalable AI Soccer Video Analysis System

Deep research report — single fixed wide-camera, fully automated stats

Prepared for Adrian · May 2026

Executive summary

A fully automated pipeline that turns single fixed wide-camera match footage into stats is technically feasible today — the entire chain (detect → track → identify → localize on pitch → derive stats) is now a single named research benchmark, **SoccerNet Game State Reconstruction (GSR)**, with public baselines you can fork [15]. The pieces that have become reliable: player detection (~0.90–0.97 mAP@0.5 when fine-tuned), multi-object tracking (HOTA ~55–70 on soccer), team assignment, and one-time pitch calibration for a fixed camera. The pieces that remain genuinely hard at amateur resolution: **ball tracking** (small, fast, occluded — the single biggest failure point) and **per-player jersey-number identification** (far-side numbers are frequently illegible).

A realistic target for a solo developer / small team: deliver reliable team-level and spatial stats (possession, territory, heatmaps, distance/speed, formations, pass networks) automatically, while treating precise individual identification and fine-grained event values (xG, packing) as best-effort with confidence flags. The fastest path to production is to fork **Roboflow Sports** (Apache-2.0) for the CV scaffold, fine-tune detection on your own footage plus permissively licensed data, and build a queue-based GPU job pipeline (serverless GPU like Modal/RunPod, or AWS g5 spot) that stores intermediate artifacts so analytics can be recomputed cheaply.

Three things to internalize before building:

1. **Garbage in, garbage out.** Every tactical metric (pitch control, xT, xG) is only as good as the upstream tracking + homography. Invest there first.
2. **The ball is your hardest problem**, not the players. Budget dedicated modeling (temporal heatmap models + trajectory interpolation) for it.
3. **Licensing matters for a commercial product.** SoccerNet is research/non-commercial only [30]; for commercial training rely on your own footage plus permissively licensed datasets.

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1. System overview — the canonical pipeline

The end-to-end flow that essentially every system (Veo, Pixellot, Track160, SoccerNet GSR) implements:

```
Video ingest → Frame decode → Player/ball detection → Multi-object tracking
   → Team assignment + jersey-number ID → Pitch calibration (pixel→metres)
   → Event detection (passes/shots/goals) → Tactical metrics → Storage/UI
```

The single fixed wide camera both helps and hurts. **It helps** because the homography is essentially constant — you calibrate once and reuse it, with no per-frame camera-motion tracking, and a static background aids both line detection and background subtraction [1][2]. **It hurts** because players (especially on the far touchline) are small, the ball is often under ~10 px, and lens distortion on a wide/fisheye capture must be modeled or far-field positions warp badly [2][9].

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2. Player & ball detection and tracking

2.1 Detection

The YOLO family dominates soccer detection. A fine-tuned **YOLOv8** is the de-facto baseline and reaches roughly **0.90–0.97 mAP@0.5 for players**, but ball detection lags badly (~0.6–0.8 mAP@0.5) because of the small object size [1][2]. YOLOv9/v10/**YOLOv11** (Ultralytics, 2024) give incremental small-object gains via improved feature pyramids, though soccer-specific head-to-head benchmarks are sparse and mostly vendor claims [3]. **RT-DETR** (transformer-based, real-time) is competitive and handles occlusion slightly better but is heavier.

The key trick for the small ball / small far-side players on a wide camera is **SAHI (Slicing Aided Hyper Inference)** — tile each frame and run detection per tile, then merge. It materially improves tiny-object recall at the cost of extra compute [12 / arXiv:2202.06934].

2.2 Multi-object tracking

Tracker	Year	Strengths	Notes for fixed-camera soccer
ByteTrack	2021 (ECCV'22)	Associates high *and* low-confidence boxes; fast	The default soccer baseline; strong through partial occlusion [arXiv:2110.06864]
BoT-SORT	2022	Adds appearance (Re-ID) + camera-motion comp.	Appearance helps ID switches; CMC less needed on a fixed cam [arXiv:2206.14651]
OC-SORT	2022 (CVPR'23)	Observation-centric; non-linear motion/ occlusion	Good for fast direction changes, no appearance model [arXiv: 2203.14360]
StrongSORT	2022	Appearance-based, accurate, heavier	Use if compute allows and ID stability matters

On the **SoccerNet-Tracking** benchmark (12 matches, 200+ annotated 30-second clips) [arXiv:2204.06918], baselines land around **HOTA in the 55–70 range, IDF1 ~70–80**. Player tracking is "largely solved" at broadcast quality; the residual difficulty is maintaining identity through occlusion clusters near the ball and when players leave/re-enter at the touchline — which is exactly where appearance Re-ID and the low-confidence association in ByteTrack earn their keep [7][8]. A convenient implementation route is **boxmot** (github.com/mikel-brostrom/boxmot), which plugs ByteTrack/BoT-SORT/OC-SORT/StrongSORT into any YOLO detector [OSS-6].

2.3 Ball tracking — the hard problem

The ball is the #1 failure point: motion blur, occlusion by players, leaving frame, and confusion with white lines/objects. Generic detectors miss it in many frames. The established remedy is a **temporal heatmap model** — **TrackNet** (Huang et al., 2019, IEEE AVSS) and **TrackNetV2** (Sun et al., 2020) — which consume several consecutive frames to exploit ball motion and predict a heatmap of ball position, designed precisely for tiny fast balls [arXiv:1907.03698; TrackNetV2 via its GitHub release — note the commonly cited arXiv ID is unverified]. TrackNetV3 (2023) adds rectification and motion attention. In practice you combine: a detector + temporal heatmap model + **Kalman / parabolic trajectory interpolation** to fill gaps during occlusion and airborne phases.

> **Reality check:** clean ball-tracking accuracy numbers are rarely published because they vary enormously with resolution. Expect this to be the component you iterate on the longest, and the one most improved by capture quality (higher resolution, higher frame rate).

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3. Player identification — jersey numbers, re-ID, team assignment

This is where amateur single-camera footage hits its hardest ceiling for *individual* stats. Three layers combine:

Team assignment (reliable). Crop each player, extract a color/deep embedding, cluster into two teams plus goalkeeper(s) and referee. The Roboflow Sports approach uses **SigLIP embeddings → UMAP → K-means**, which is robust when kits are visually distinct and degrades when colors or lighting are similar [6]. This part generally works well.

Jersey number recognition (partial). The **SoccerNet Jersey Number Recognition** challenge frames this as a *tracklet-level* problem: run scene-text recognition per frame (modern transformer OCR like PARSeq/TrOCR beats older CRNN), then aggregate across the whole tracklet by confidence-weighted voting [1][2]. The crucial insight — captured in Koshkina & Elder's "A General Framework for Jersey

Number Recognition in Sports Video" (CVPRW 2024, arXiv:2405.13896) — is that you only need a handful of legible frames per tracklet, so temporal aggregation hugely outperforms per-frame OCR. Reported accuracy is ~85–92% on *legible* tracklets but much lower once "illegible" cases (player facing away, occluded, blurred, far side) are included [2][3]. A large fraction of tracklets must simply be predicted as "unknown."

Re-identification (hard in soccer). Standard person Re-ID fails because teammates wear identical kits — appearance embeddings cannot separate them [4][5]. SoccerNet Re-ID baselines sit well below street-person Re-ID datasets. The conclusion: **appearance Re-ID alone cannot identify individuals on a team.** Robust identity comes only from combining tracking (short-term continuity) + jersey number (aggregated over the tracklet) + team assignment + spatial constraints — exactly the **SoccerNet GSR** formulation [15].

> **What's realistic on your footage:** reliable team assignment and tracking IDs; *partial* individual jersey-number ID (good for near-side / larger crops, frequently "unknown" for the far touchline). Full season-long per-player attribution typically needs either higher-resolution/zoomed capture, or — as Trace does commercially — a player-worn tag as a shortcut [27].

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4. Pitch calibration & homography (pixel → metres)

Mapping pixels to the standard 105×68 m pitch model is **mandatory** for any spatial stat (positions, distance, speed, heatmaps, formations). For a fixed camera this is a one-time (or stable-window-averaged) computation — a major simplification over broadcast pipelines [1][2].

Methods, in rough order of robustness for soccer:

- **Line/keypoint-based registration** — detect pitch lines and their intersections (known geometry), solve the homography from correspondences. Most robust because pitch markings are standardized [1][3].
- **Learned keypoint detection + fit** — a model segments lines/keypoints, then you fit camera parameters. This is the SoccerNet calibration baseline approach and what Roboflow Sports uses for its radar/minimap view [4][11].
- **Optimization-based** — **TVCalib** (Theiner & Ewerth, WACV'23, arXiv:2207.11709) optimizes over pitch segments; the **"No Bells, Just Whistles"** group (CVPRW 2024, arXiv:2404.08401) leverages geometric properties (PnLCalib is their points-and-lines follow-up). Earlier foundational work: Jiang et al. "Optimizing Through Learned Errors..." (WACV'20, arXiv:1909.08034) and Nie et al. "A Robust and Efficient Framework for Sports-Field Registration" (WACV'21, arXiv:2103.13868).

Single-fixed-camera specifics. Pro: calibrate once, static background aids line detection. Cons: wide-angle/fisheye **radial distortion must be modeled** or the far field warps; and the far side of the pitch has very few pixels per metre, so positional error — and especially its derivative, speed — grows with distance from the camera. **Kalman smoothing before computing speed/distance is essential.** A few pixels of reprojection error can translate to roughly 0.5–2 m of positional error depending on distance [2][8].

Open-source starting points: `SoccerNet/sn-calibration`, Roboflow Sports keypoint pitch detection, and `MM4SPA/tvcalib`.

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5. Event detection & tactical metrics

5.1 Events: two paradigms

(a) Action spotting on raw video. The **SoccerNet Action Spotting** challenge spots 17 classes (goal, card, sub, corner, foul…) with precise temporal location; **SoccerNet Ball Action Spotting** (2023+) targets finer ball-centric actions (pass, drive, header, shot) — closer to what you want. Methods evolved NetVLAD++ (Giancola & Ghanem 2021, arXiv:2104.06779) → temporal transformers (T-DEED, E2E-Spot, "COMEDIAN"), with Average-mAP climbing from the ~50s to ~70s over 2021–2024 [1][3]. This paradigm is best for goals, cards, and set pieces.

(b) Events derived from tracking geometry. Assign the ball to the nearest player meeting proximity + velocity criteria; possession changes define passes; same-team transfer = completed pass; a trajectory toward goal = shot. This is interpretable and standard in analytics pipelines, and is the better route for possession/passing stats [4][5].

For a production system you'll likely **fuse both** — action spotting for discrete match events, tracking-derived logic for continuous possession/passing.

5.2 Tactical metrics (from tracking + calibrated positions)

Metric	Source	Needs
Pitch control	Spearman 2017/2018	Player positions + velocities
Space value / EPV	Fernández & Bornn "Wide Open Spaces" 2018; EPV 2019	Positions, possession value model
xT (Expected Threat)	Karun Singh 2019 (karun.in/blog/expected-threat.html)	Grid + ball location over possessions
VAEP	Decroos et al., KDD 2019 (arXiv:1802.07127)	Event stream
xG	StatsBomb open model / open-data	Accurate shot location + context
Formation detection	Bialkowski et al. 2014	Average positions + role assignment
Pressing (PPDA)	Industry standard	Events / tracking

What's realistically computable from amateur single-camera footage: possession %, territory/heatmaps, distance covered, sprint counts, team shape/formation, pitch-control surfaces, and pass networks (where player ID succeeds). **What degrades or fails:** precise xG (needs accurate shot location, body part, context), packing/line-breaking (needs precise positions), and anything off-ball that leaves the frame. Every one of these inherits the noise of the tracking + homography upstream [5][6][11].

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6. Datasets & benchmarks

SoccerNet family (soccer-net.org) — the dominant resource, built on 500 broadcast matches:

- **SoccerNet v1** (2018, arXiv:1804.04527) — 500 matches (~764 h), action spotting (3 classes).
- **SoccerNet-v2** (2021, arXiv:2011.13367) — 17 action classes (~110k annotations), camera-shot segmentation, replay grounding.
- **SoccerNet-v3** (Nature Scientific Data 2022, DOI 10.1038/s41597-022-01469-1) — multi-view spatial localization, bounding boxes, lines, re-ID links.
- **Challenge tracks:** Action Spotting, Ball Action Spotting, Camera Calibration, Player Tracking/MOT (arXiv:2204.06918), Re-Identification, Jersey Number Recognition, Dense Video Captioning, **Game State Reconstruction** (CVPRW 2024, arXiv:2404.11335), Monocular Depth.

Most relevant to a fixed wide camera:

- **SoccerTrack** (CVPRW/CVSports 2022, CVF Open Access — *no arXiv preprint*; code now at github.com/AtomScott/**SportsLabKit**, the old SoccerTrack repo redirects there) — **fish-eye wide-view + drone footage with GNSS ground truth**, the closest public match to your capture setup. SoccerTrack v2 (2025) data is CC BY 4.0 (commercial-friendly).
- **Roboflow Universe** soccer/football detection datasets — YOLO-ready, varied (often permissive) licenses; the most practical starting point for fine-tuning.
- **DFL Bundesliga Data Shootout** (Kaggle 2022) — event detection.
- **SoccerDB** (arXiv:1912.04465) — detection + action.
- **WorldCup 2014** — standard field-registration benchmark.
- **StatsBomb Open Data** (github.com/statsbomb/open-data) — free *event* data (with 360 freeze-frames) to validate your analytics layer.

> ⚠ **Licensing:** SoccerNet video is **research/non-commercial only** and gated behind an access agreement; the dev-kit code is MIT [30]. For a commercial product, train on **your own collected footage** plus permissively licensed Roboflow sets, and use SoccerNet only for prototyping/benchmarking.

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7. Competitive landscape (build vs. buy)

Closest to your setup — grassroots camera + auto-analysis:

- **Veo** (Denmark) — fixed dual-lens 4K camera; AI auto-records and digitally follows the ball with no operator; expanding into analytics, heatmaps, player stats. Raised an **$80M Series C in June 2022, led by ATP** (with Ventech, Seed Capital). Pricing is indicative and quote-based: roughly **~$1,299 hardware + ~$800–1,300/yr subscription** (Essential→Pro tiers), varies by market — verify current rates [1][25][26].
- **Pixellot** — AI panoramic (stitched) single-unit capture + auto production and stats; powers many federations and the NFHS Network in the US [2].
- **Trace (TraceUp / Once Sport)** — autonomous camera **plus a player-worn GPS tag** to identify players and auto-cut per-player clips/stats. A pragmatic hybrid that sidesteps the visual-ID ceiling [3][27].
- Also: **Spiideo** (cloud multi-cam + Perform AI analytics), **Bepro**, **Reeplayer**, **XbotGo** (robotic phone mount).

Pro tracking/analytics providers:

- **Hudl** — dominant; acquired **Wyscout** and **StatsBomb** (2024), spanning grassroots→pro [6][23].
- **Stats Perform (Opta)** — leading event data; "Opta Vision" fuses event + tracking.
- **Genius Sports / Second Spectrum** — Genius acquired Second Spectrum (2021); in-stadium multi-camera optical tracking (official for several leagues).
- **SkillCorner** — **broadcast-video** CV tracking (no in-stadium cameras) → physical + tactical data; methodologically the closest large-scale analog to a CV-only approach [9].
- **Sportlogiq**, **PFF FC**, **Track160** (markerless single-/multi-camera CV → 3D pose + tracking), **Tracab/ChyronHego** (optical multi-camera pioneer).

Takeaway: single-/fixed-camera CV → stats is commercially proven, but every serious player mitigates the resolution ceiling somehow — wide multi-lens capture (Veo, Pixellot), a worn tag (Trace), or operating on higher-quality broadcast feeds (SkillCorner). Plan a similar mitigation.

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8. Production architecture & MLOps

8.1 Open-source foundation

- **Roboflow Sports** (github.com/roboflow/sports, **Apache-2.0**) — the reference scaffold: player detection, team clustering (SigLIP+UMAP+KMeans), pitch keypoint detection + homography/radar view, ball tracking. Best place to start [OSS-1].
- **Roboflow Supervision** — detections, ByteTrack integration, annotators, zones [OSS-2].
- **SoccerNet dev-kit** — baselines + dataloaders for every track, including GSR [OSS-3].
- **boxmot** — pluggable MOT trackers [OSS-6]. **narya** is an older full pipeline reference (limited maintenance).

8.2 Inference & video pipeline

The cost driver is GPU video processing, so optimize it:

- **Decode on GPU** (NVDEC via NVIDIA DeepStream / PyNvVideoCodec) to avoid a CPU bottleneck on 90-minute 4K; `decord` / PyAV for simpler Python access.
- **Compile models with TensorRT** (FP16, or INT8 with calibration) for ~2–5× speedup; export path is YOLO → ONNX → TensorRT.
- **Serve with NVIDIA Triton** (dynamic batching, multi-model, GPU sharing), or use **DeepStream** end-to-end for many concurrent streams at low cost-per-stream.
- **Adaptive frame sampling:** don't run every model at full frame rate. Run detection/tracking at ~5–15 fps for positions; give the **ball model a higher cadence**; run action spotting on its own schedule.

Decouple stages with queues: `ingest → decode → batched detection → tracking → calibration → analytics → store`.

8.3 Scale & cost

- **GPU options:** AWS **g4dn.xlarge** (T4, ≈ $0.526/hr) is the cheapest inference workhorse; **g5.xlarge** (A10G, ≈ $1.006/hr) is faster; **spot** cuts ~60–70% [28]. **Serverless GPU** (Modal, RunPod, Replicate) bills per-second and scales to zero — ideal for spiky "match uploaded" workloads, avoiding idle-GPU cost.
- **Rough order-of-magnitude** to process one 90-minute match with a TensorRT detection+tracking stack at ~real-time on a T4/A10G: ~1–3 GPU-hours → ~$0.50–$3 of GPU compute plus storage/egress. Adding heavier per-frame models (ball heatmap, action spotting, re-ID) raises this. *Treat as a planning estimate, not a quote — it depends heavily on model stack and fps* [28].
- **Storage:** S3 for raw video (4K/90 min = several to tens of GB) with lifecycle tiering; structured outputs in Postgres.
- **Critical pattern: persist intermediate artifacts** (detections, tracks, homography). The CV stage is the expensive part — store its output so you can recompute *analytics* (new metrics, bug fixes) without re-running CV.

8.4 MLOps & the data flywheel

- **Job orchestration:** Celery+Redis / SQS, or native queues in Modal/RunPod; one match = one job, optionally fanned out by time segment for parallelism. Make jobs idempotent and resumable (checkpoint per segment).
- **Versioning:** MLflow or W&B for models; DVC or Roboflow for datasets.
- **Evaluation pipeline:** keep a held-out labeled set; track HOTA/IDF1/mAP per model version; regression-test before deploy.
- **Automated QA gates per job:** % of frames with ball detected, number of ID switches, calibration reprojection error → auto-flag low-quality outputs for review instead of shipping bad stats silently.
- **Active learning loop:** route low-confidence detections and broken tracklets to annotation (CVAT/Label Studio/Roboflow) → retrain. This is how you climb past the prototype ceiling on *your* footage distribution.

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9. Recommended roadmap (prototype → production)

1. **Fork Roboflow Sports**; reproduce detection + team clustering + pitch homography on a few of your own matches.
2. **Fine-tune detection** on your footage + Roboflow Universe + SoccerTrack (wide/fisheye matches your camera). Lock in calibration with a fisheye/radial-distortion model.
3. **Harden tracking** (ByteTrack → BoT-SORT with appearance) and **invest in the ball** (TrackNet-style temporal model + trajectory interpolation). Measure HOTA/IDF1 and ball-detection rate as your north-star CV metrics.
4. **Layer identity** (team assignment first, jersey-number aggregation second) with explicit "unknown" handling and confidence scores.
5. **Add analytics** from calibrated tracks: possession, heatmaps, distance/speed (Kalman-smoothed), formations, pass networks; fuse with action spotting for discrete events. Validate metric logic against StatsBomb open data.
6. **Productionize:** TensorRT + Triton/DeepStream, GPU-decode, queue + serverless/spot GPU workers, artifact persistence, automated QA gates, and a labeling→retraining flywheel.
7. **Decide your resolution-ceiling mitigation** (higher-res capture, multi-lens, or — for guaranteed per-player stats — an optional worn tag à la Trace).

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10. Key open questions / confidence flags

- **Ball-tracking accuracy** has no clean published benchmark for amateur footage — you must measure it on your own data; it is the dominant risk.
- **Per-player ID coverage** on a single wide camera will be partial; quantify the legible fraction early to set product expectations.
- **Cost estimates and cloud prices** (§8.3) change frequently — re-check before committing.
- **Veo funding/pricing, Hudl–StatsBomb acquisition month, exact AWS prices** are approximate and time-sensitive.
- Citations were adversarially fact-checked; a few works have **no arXiv preprint** and are cited via their official venue (TrackNetV2 → IEEE ICPAI 2020; SoccerNet-v3 → Nature Scientific Data; Nie et al. and SoccerTrack → CVF Open Access). PnLCalib and "No Bells, Just Whistles" share arXiv:2404.08401 (same evolving work).

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